Waterloo Road and the Clock in the Streets

Battle street names and the growth of British towns, from OpenStreetMap

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DOI: https://doi.org/10.5281/zenodo.24086515

2026

Abstract

Nineteenth-century builders often named new streets after recent victories, so battle names may record when a street was laid out. I match 6,885 OpenStreetMap highways in the United Kingdom against a table of 111 battles, join the pieces of each street into 3,352 streets, and set aside streets named after battles before 1793 (1,008) and names that usually refer to something else (654: Nile, Quebec, Kimberley and others). Of 1,690 streets named after 54 battles from 1793 to 1944, Waterloo (367), Alma (340) and Trafalgar (241) lead; the four Crimean battles gave 425 streets, more than both world wars together (315), and world-war names are much more often modern estate names (39.4% Close, Drive, Way and the like, against 12.9% for 1854 to 1902). Within towns, streets named after later battles lie farther from the centre: the mean within-town Spearman correlation of battle date and distance is 0.25 across 75 towns (53 positive, 19 negative; permutation $p < 0.001$, or $p < 0.01$ with clusters of same-war streets counted once). Ranked within their towns, Napoleonic streets sit at 0.54, Crimean at 0.58 and Boer War streets at 0.68 of the way from centre to edge. The clock is real but coarse: it separates wars half a century apart, not neighbouring ones.

Keywords: street names; commemoration; urban growth; toponymy; battles; OpenStreetMap

1 Introduction

Street names are a public record of what a society chose to commemorate (Azaryahu, 1996; Alderman, 2000; Rose-Redwood et al., 2010). In Victorian Britain one common choice was the latest victory: Alma Road and Inkerman Street after 1854, Ladysmith Road and Mafeking Street in 1900. If developers named streets as they were laid out, and if towns grew outwards, battle names should form a crude clock in the street plan, with older victories near the centre and newer ones farther out, echoing the rings of growth that urban morphologists have traced in British towns (Conzen, 1960; Whitehand, 1972). This paper asks whether the clock is there, using an open map of every street in the United Kingdom.

2 Data

Streets. OpenStreetMap ways with a `highway` tag in the United Kingdom whose names contain a battle name, queried with QLever (Bast and Buchhold, 2017): 6,885 ways with highway type, name and first coordinate. OpenStreetMap's coverage of names is good in towns but uneven (Haklay, 2010). **Towns.** Wikidata settlements typed as towns or cities with a population (CC0). **Counties.** ONS boundaries.

Battle table. Each battle has the date of the battle, or for a siege the date of its relief or fall (Table 1). Battles before 1793 are kept but set aside from the clock, since a street named Naseby or

Agincourt was named centuries after the event. Names that usually refer to a place, a person or a colour are flagged as ambiguous and set aside: Aboukir, Alexandria, Cabul, Copenhagen, Crimea, Delhi, Dunkirk, Falklands, Gibraltar, Kandahar, Kimberley, Magenta, Minorca, Nile, Portobello, Quebec, Scinde, St Vincent and Vigo. The Nile, for example, is far more often the river; London's Copenhagen Street takes its name from the area around Copenhagen House, a seventeenth-century house far older than the battle; Dunkirk was an English place name long before 1940; Magenta is a colour, itself named after the battle.

3 Method

Matching and joining. Names were matched to the battle table as whole words; names of twin towns (Villiers-sur-Marne Avenue) were excluded. Pieces with the same normalised name were joined when their first points chain within 600 m, and pieces of through roads (primary to unclassified, tracks, cycle paths) within 1.5 km, which joined 3,509 pieces into others; 11 pairs of same-named streets within 1.5 km remain separate, since they may be distinct streets. The full list of names for each battle was read; 9 were removed by hand (`code/overrides.csv`): non-streets such as pothole markers, moorings and railway sidings, Alma Martin Way, De-Verdun Avenue, and Arras Hill near Market Weighton, an old Yorkshire place name. 32 streets named after battles from 1793 on lie within 3 km of a Wikidata place containing the battle's name (Waterloo in London and on Merseyside, Inkerman in County Durham, Camperdown in Dundee) and are set aside as probably named after the place.

Towns and distance. Each street is assigned to the nearest town of at least 20,000 people within 10 km, with all of Greater London counted as London, and its distance from the town's Wikidata point is measured. **Street words.** The last word of the name is classed as traditional (Street, Road, Terrace, Place, Row, Square, Grove, Crescent, Lane, Villas and similar), modern (Close, Drive, Way, Court, Mews, Rise, View, Gardens and similar) or other.

The clock test. The main test uses streets named after battles from 1793 on, with a traditional street word, not ambiguous or place-named, in a town. For each town with at least four such streets and two distinct battle years, I compute Spearman's rank correlation between battle year and distance from the centre. The statistic is the mean correlation across towns; its null distribution comes from 4,000 permutations of battle years within towns, which keeps each town's set of distances and dates. A sign test on the number of positive and negative towns is reported alongside. As a second view, each street's distance is ranked within its town (towns with three or more streets) and the mean rank percentile is computed for each war with a bootstrap interval; with no clock every war would sit at 0.5.

Table 1: The battles found in street names: date used, war, status (old: before 1793, set aside; amb.: ambiguous name, set aside) and number of streets.

Battle	Date	War	Status	Streets
Armada	1588-08-08	Spanish Armada	old	28
Bannockburn	1314-06-24	Scottish independence	old	12
Crecy	1346-08-26	Hundred Years War	old	14
Poitiers	1356-09-19	Hundred Years War	old	2
Agincourt	1415-10-25	Hundred Years War	old	27
Bosworth	1485-08-22	Wars of the Roses	old	73
Flodden	1513-09-09	Anglo-Scottish wars	old	9
Edgehill	1642-10-23	English Civil War	old	68
Marston Moor	1644-07-02	English Civil War	old	5

Table 1: The battles found in street names: date used, war, status (old: before 1793, set aside; amb.: ambiguous name, set aside) and number of streets.

Battle	Date	War	Status	Streets
Naseby	1645-06-14	English Civil War	old	78
Sedgemoor	1685-07-06	Monmouth Rebellion	old	40
Boyne	1690-07-11	Williamite War	old	31
Aughrim	1691-07-22	Williamite War	old	10
Vigo	1702-10-23	Spanish Succession	amb.	22
Blenheim	1704-08-13	Spanish Succession	old	478
Gibraltar	1704-08-04	Spanish Succession	amb.	52
Ramillies	1706-05-23	Spanish Succession	old	19
Almanza	1707-04-25	Spanish Succession	old	1
Oudenarde	1708-07-11	Spanish Succession	old	1
Malplaquet	1709-09-11	Spanish Succession	old	1
Portobello	1739-11-22	Jenkins' Ear	amb.	32
Dettingen	1743-06-27	Austrian Succession	old	6
Fontenoy	1745-05-11	Austrian Succession	old	2
Culloden	1746-04-16	Jacobite rising	old	28
Minorca	1756-05-20	Seven Years War	amb.	16
Plassey	1757-06-23	Seven Years War	old	11
Louisburg	1758-07-26	Seven Years War	old	1
Minden	1759-08-01	Seven Years War	old	34
Quebec	1759-09-13	Seven Years War	amb.	72
Quiberon	1759-11-20	Seven Years War	old	1
Belle Isle	1761-06-07	Seven Years War	old	15
Havannah	1762-08-13	Seven Years War	old	13
St Vincent	1797-02-14	French Revolutionary Wars	amb.	59
Camperdown	1797-10-11	French Revolutionary Wars		27
Nile	1798-08-01	French Revolutionary Wars	amb.	54
Aboukir	1798-08-01	French Revolutionary Wars	amb.	2
Copenhagen	1801-04-02	French Revolutionary Wars	amb.	22
Alexandria	1801-03-21	French Revolutionary Wars	amb.	24
Trafalgar	1805-10-21	Napoleonic Wars		241
Maida	1806-07-04	Napoleonic Wars		23
Vimeiro	1808-08-21	Peninsular War		1
Corunna	1809-01-16	Peninsular War		21
Talavera	1809-07-28	Peninsular War		11
Salamanca	1812-07-22	Peninsular War		14
Vittoria	1813-06-21	Peninsular War		8
Waterloo	1815-06-18	Napoleonic Wars		367
Algiers	1816-08-27	Bombardment of Algiers		4
Navarino	1827-10-20	Greek War of Independence		4
Cabul	1842-09-15	First Anglo-Afghan War	amb.	3
Meeanee	1843-02-17	Conquest of Sind		2
Scinde	1843-02-17	Conquest of Sind	amb.	1
Moodkee	1845-12-18	First Anglo-Sikh War		1
Ferozeshah	1845-12-21	First Anglo-Sikh War		1
Aliwal	1846-01-28	First Anglo-Sikh War		3
Sobraon	1846-02-10	First Anglo-Sikh War		2
Goojerat	1849-02-21	Second Anglo-Sikh War		1
Alma	1854-09-20	Crimean War		340
Balaclava	1854-10-25	Crimean War		22
Inkerman	1854-11-05	Crimean War		57
Sebastopol	1855-09-09	Crimean War		6
Crimea	1855-09-09	Crimean War	amb.	14

Table 1: The battles found in street names: date used, war, status (old: before 1793, set aside; amb.: ambiguous name, set aside) and number of streets.

Battle	Date	War	Status	Streets
Delhi	1857-09-20	Indian Rebellion	amb.	18
Cawnpore	1857-07-16	Indian Rebellion		5
Lucknow	1857-11-16	Indian Rebellion		21
Magenta	1859-06-04	Second Italian War	amb.	11
Ulundi	1879-07-04	Anglo-Zulu War		3
Kandahar	1880-09-01	Second Anglo-Afghan War	amb.	4
Majuba	1881-02-27	First Boer War		3
Tel el Kebir	1882-09-13	Anglo-Egyptian War		1
Khartoum	1885-01-26	Mahdist War		14
Dargai	1897-10-20	Tirah Campaign		3
Omdurman	1898-09-02	Mahdist War		2
Modder River	1899-11-28	Second Boer War		5
Colenso	1899-12-15	Second Boer War		33
Spion Kop	1900-01-24	Second Boer War		9
Kimberley	1900-02-15	Second Boer War	amb.	173
Tugela	1900-02-27	Second Boer War		4
Ladysmith	1900-02-28	Second Boer War		73
Mafeking	1900-05-17	Second Boer War		43
Mons	1914-08-23	First World War		30
Marne	1914-09-06	First World War		30
Ypres	1914-10-19	First World War		13
Gallipoli	1915-04-25	First World War		5
Anzac	1915-04-25	First World War		7
Verdun	1916-02-21	First World War		18
Jutland	1916-05-31	First World War		34
Somme	1916-07-01	First World War		12
Arras	1917-04-09	First World War		18
Vimy	1917-04-09	First World War		12
Messines	1917-06-07	First World War		6
Cambrai	1917-11-20	First World War		21
Dunkirk	1940-05-26	Second World War	amb.	55
Tobruk	1941-04-10	Second World War		9
Alamein	1942-10-23	Second World War		35
Anzio	1944-01-22	Second World War		12
Cassino	1944-01-17	Second World War		7
Kohima	1944-04-04	Second World War		8
Imphal	1944-03-08	Second World War		1
Arnhem	1944-09-17	Second World War		37
Falklands	1982-04-02	Falklands War	amb.	20

4 Results

4.1 Which battles

The 1,690 streets after battles from 1793 on are dominated by three names: Waterloo (367), Alma (340) and Trafalgar (241), followed by Ladysmith (73), Inkerman (57) and Mafeking (43) (Figure 1). Before 1793, Blenheim is the commonest battle name of all (478), though often through Blenheim Palace. The names come in bursts: after Waterloo the battles of 1816 to 1853 gave only 18 streets, while the Crimean War gave 425 in two years. The First World War gave 206 and the

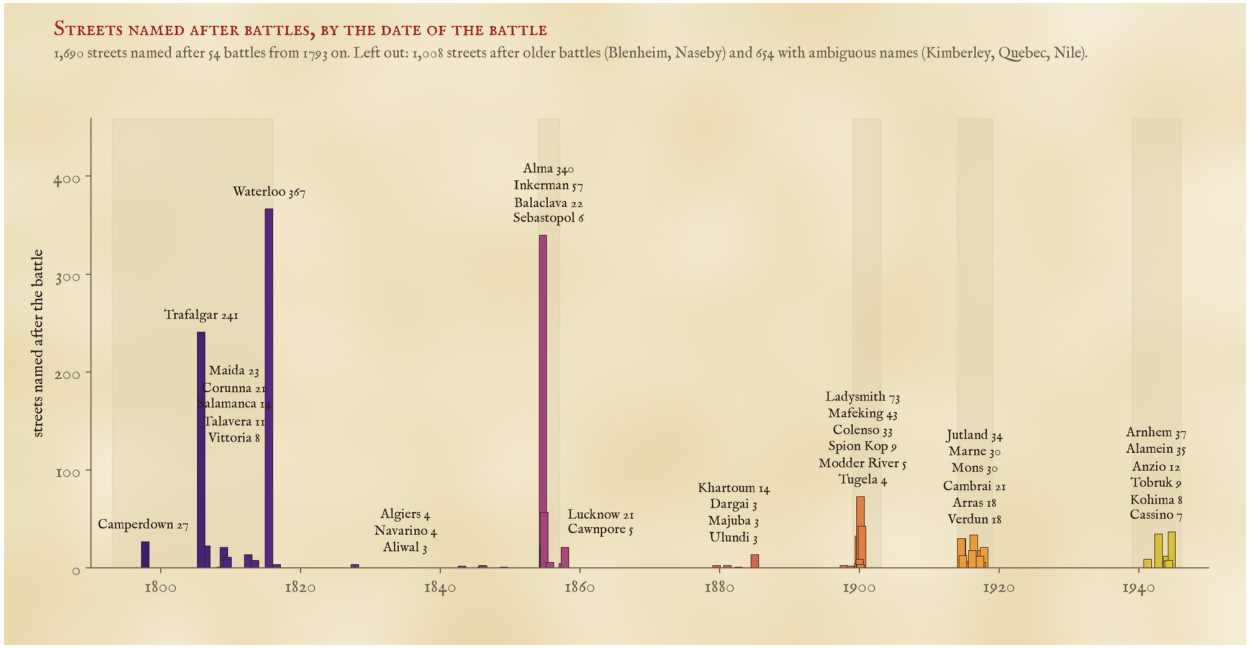


Figure 1: Streets named after each battle from 1793 on, by the date of the battle.

Second 109. By county, Greater London (126), Greater Manchester (91) and West Yorkshire (82) have the most; the Alma is the commonest battle in 18 of 46 English counties, and Trafalgar leads in naval Hampshire and Devon (Figure 2).

4.2 Named at the time, or later?

For battles from 1854 to 1902, 78.9% of streets end in a traditional word and 12.9% in a modern one. For the world-war battles the modern share is 39.4% (Figure 3), consistent with many of those names being chosen as estate themes decades after the war. Waterloo and Trafalgar also kept being reused: 33.2% of Trafalgar streets and 19.9% of Waterloo streets have a modern street word. This is why the main test uses traditional street words only.

4.3 The clock

Across 75 towns with four or more streets (549 streets), the mean within-town correlation of battle year and distance is 0.25 (median 0.32); 53 towns are positive and 19 negative (sign test $p = 0.00004$). None of 4,000 within-town permutations reached the observed mean; the 97.5th percentile of the null is 0.11 (Figure 4). The result holds without London (mean 0.26, 74 towns), for towns with six or more streets (0.25, 36 towns), and for battles up to 1902 only (0.18, 62 towns, 4 of 4,000 permutations as high). With all street words it is weaker (0.14, 110 towns, 6 of 4,000 permutations as high), as expected if modern names were given late. The permutation test treats streets as independent, but streets of one war often stand together and were probably named together. Counting each cluster of same-war streets within 500 m of each other in a town as one unit (792 units), the mean correlation falls to 0.16 across 51 towns (32 positive, 17 negative), with 27 of 4,000 permutations as high ($p = 0.007$). The clock survives, but the street-level test overstates its precision.

Within their towns, Napoleonic streets sit at a mean rank of 0.54 (interval 0.51 to 0.58), Crimean at 0.58 (0.54 to 0.62), Boer War at 0.68 (0.63 to 0.72) and First World War at 0.66; median distances

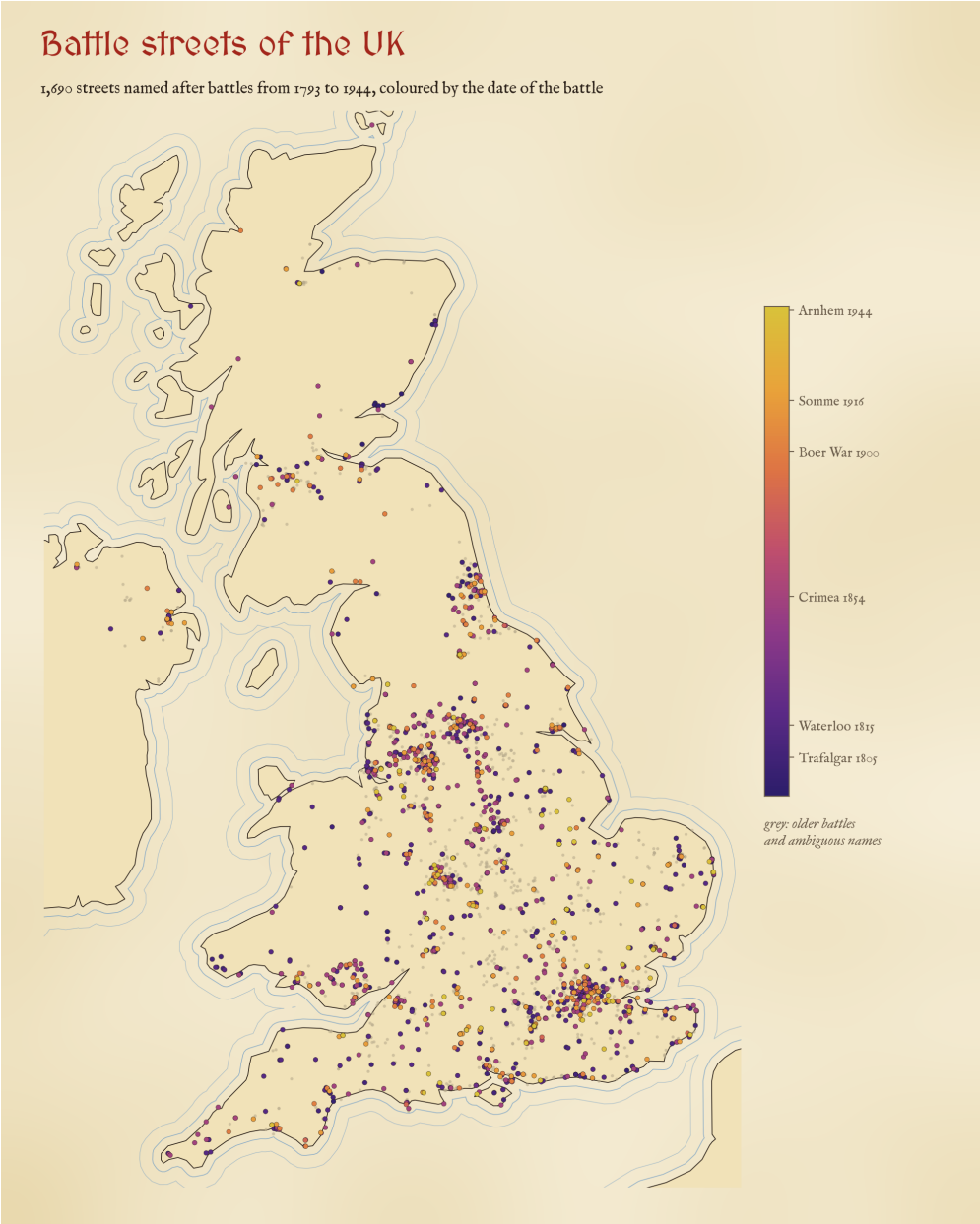


Figure 2: Battle streets in the United Kingdom, coloured by the date of the battle; grey: older battles and ambiguous names.

are 1.45, 2.01, 2.36 and 3.00 km.

4.4 A coarse clock

Pairwise, the clock separates wars far apart but not neighbours. Napoleonic streets are nearer the centre than Boer War streets in 30 of 40 towns with both, Crimean than Boer War in 23 of 38, but Napoleonic than Crimean in only 47 of 92, and Boer War than First World War in 8 of 17. Individual towns vary (Figure 5): Bristol (0.66), Glasgow (0.75) and Hull (0.84) keep good time, London barely (0.12), and Preston (-0.33) and Lincoln (-0.84) run backwards.

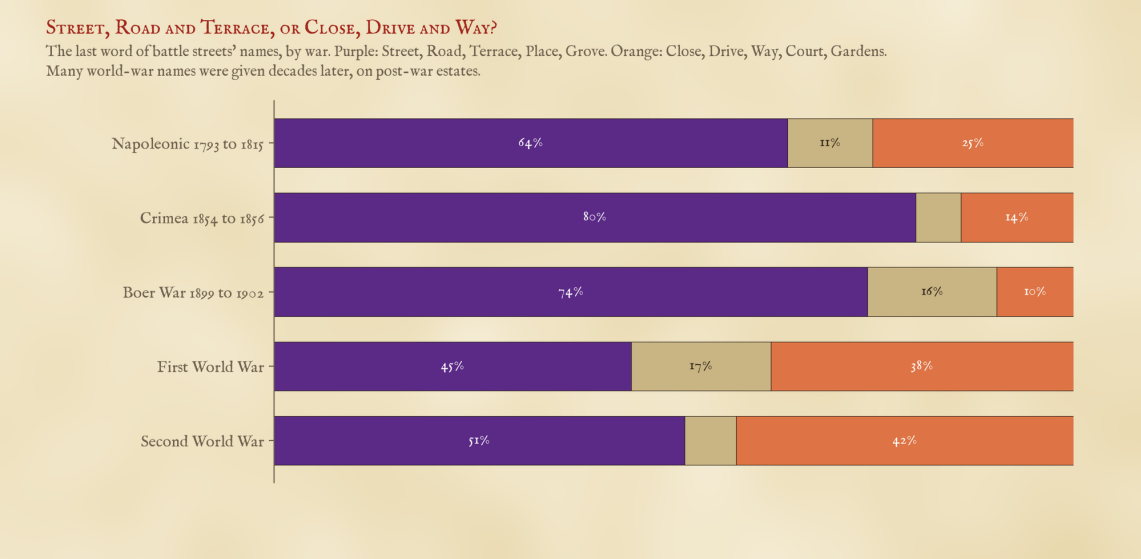


Figure 3: The last word of battle street names, by war.

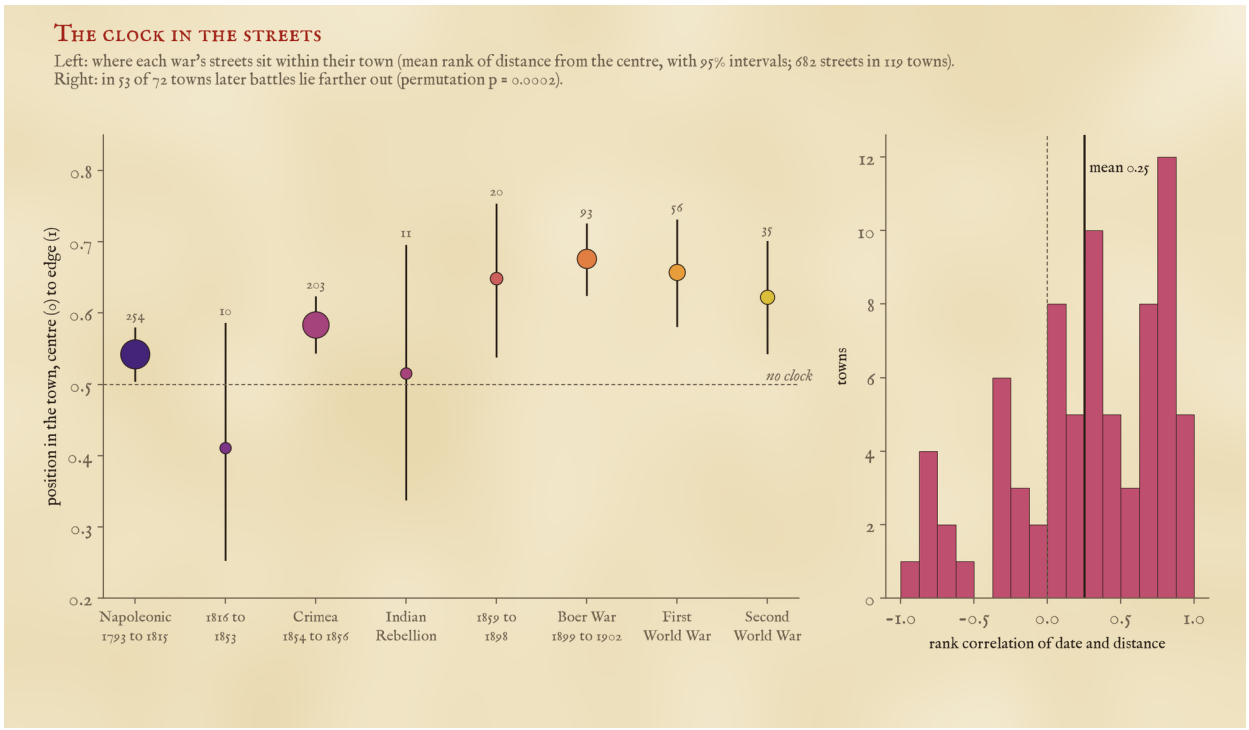


Figure 4: Left: mean within-town rank of distance by war, with 95% bootstrap intervals. Right: within-town correlations of battle date and distance.



Figure 5: Six towns: battle streets around the centre and the median distance of each war's streets.

5 Limitations

The name dates the battle, not the street: some streets were named long after their battle, some renamed, and Waterloo and Trafalgar especially stayed in fashion. The traditional street-word filter reduces but does not remove late naming. Towns did not grow in rings: they grew along roads and railways and around industry, and a conurbation's streets are assigned to whichever town point is nearest, which compresses distances. Town centres are Wikidata points. Alma is also a woman's name: 70 of the 340 Alma streets lack a traditional street word, and 50 of these lie more than 1 km from any traditional Alma street, so some may commemorate a person or a house; even if all 50 were removed, the Crimean battles would give 375 streets, still more than both world wars. These streets are outside the main test, which uses traditional street words only. The ambiguity and place-name rules are judgements, documented in the code; a few ambiguous names may be battles and a few kept names may not be. Each street is placed at the mean of the first points of its pieces. OpenStreetMap names are incomplete in places, and the counts for many battles are small.

6 Conclusion

Battle names do carry a clock. Across British towns, streets named after later battles lie farther from the centre, clearly enough that no shuffle of the dates matched it, though counting clusters of same-war streets once makes the signal weaker. But it is a slow clock that runs fast in places: it tells the French wars from the Boer War, not Waterloo from the Alma. The bursts of names,

the Crimea above all, and the shift of world-war names onto post-war estates are as telling as the rings.

Data and code availability

The interactive map is at https://maps.tristanhotham.com/battle-streets/. Code, the battle table, the rules and hand decisions, and the derived tables are at https://github.com/Tristan-Hotham/battle-streets and archived with this paper at https://doi.org/10.5281/zenodo.24086515. The street tables are derived from OpenStreetMap (© OpenStreetMap contributors) and released under the Open Database Licence; Wikidata is CC0; county boundaries from the ONS (Open Government Licence v3.0); text and figures CC BY 4.0; code MIT.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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